1 Indistinguishability

We'll now start a major unit on *indistinguishability* and *pseudorandomness*. These concepts are a cornerstone of modern cryptography, underlying several foundational applications such as pseudorandom generators, secure encryption, "commitment" schemes, and much more.

For example, our most immediate application of indistinguishability will be to construct cryptographically strong *pseudorandom bit generators*. These are algorithms that produce many "random-looking" bits, while using very little "true" randomness. (In particular, the bits they output will necessarily be "very far" from truly random, in a statistical sense.) One easy-to-imagine application would be to use the pseudorandom bit stream as a one-time encryption pad, which would allow the shared secret key to be much smaller than the message. But what does it *mean* for a string of bits to be "random-looking"? And how can we be confident that using such bits does not introduce any unforeseen weaknesses in our system?

More generally, the motivating question for our study is:

When can two (possibly different) objects be considered *effectively the same*?

The answer:

When they can't be told apart!

Though seemingly glib, this answer encapsulates a very powerful mindset that will serve us well as we go forward.

1.1 Statistical Indistinguishability

We use probability theory to model (in)distinguishability. If two distributions are identical, then they certainly should be considered indistinguishable. We relax this condition to define *statistical* indistinguishability, for when the *statistical distance* between the two distributions is negligible. The statistical distance between two distributions X and Y over a domain Ω is defined as[1]

$$\Delta(X,Y) := \sup_{A \subseteq \Omega} |X(A) - Y(A)|,$$

where $X(A) = \sum_{w \in A} \Pr[X = w]$ is the probability that a draw from X lands in A, and likewise for $Y(A)$. We can view $A \subseteq \Omega$ as a statistical "test" that has some probability of "passing" when given an element drawn from X, or from Y; the statistical distance is essentially the maximum difference between these two probabilities, taken over all tests. Note that A and $\bar{A}$ are effectively the same test, since

$$|X(\bar{A}) - Y(\bar{A})| = |1 - X(A) - (1 - Y(A))| = |Y(A) - X(A)| = |X(A) - Y(A)|.$$

Lemma 1.1. *For distributions X, Y over a finite domain Ω,*

$$\Delta(X,Y) = \frac{1}{2} \sum_{w \in \Omega} |X(w) - Y(w)|.$$

[1] For a set S of real numbers, its *supremum* $\sup(S)$ is the least upper bound of the elements of S. The supremum can be thought of as a generalization of the maximum element to (infinite) sets where no such element may exist. For example, there is no maximum element of the set $[0,1) \subseteq \mathbb{R}$, but the supremum is 1 since it is the smallest real number that is larger than every element of $[0,1)$. When S is finite, the supremum is simply the maximum.

Proof. Let the test $A = \{w \in \Omega \ : \ X(w) > Y(w)\}$. This makes $X(A) - Y(A)$ as large as possible, so $\Delta(X,Y) = X(A) - Y(A) = \sum_{w \in A} |X(w) - Y(w)|$. As noted above, we also have $\Delta(X,Y) = Y(\bar{A}) - X(\bar{A}) = \sum_{w \in \bar{A}} |X(w) - Y(w)|$. Summing the two equations, we have $2\Delta(X,Y) = \sum_{w \in \Omega} |X(w) - Y(w)|$, as desired. □

Statistical distance is very robust, which enhances its usefulness. Using Lemma 1.1, the following facts are straightforward to prove.

Lemma 1.2. *Let f be a function (or more generally, randomized procedure) on the domain of X, Y. Then $\Delta(f(X), f(Y)) \leq \Delta(X,Y)$.*

In other words, statistical distance cannot be increased by the application of any (randomized) procedure.

Lemma 1.3. *Statistical distance is a metric; in particular, it satisfies the following three properties:*

- *Identity of indiscernibles:* $\Delta(X,Y) = 0 \iff X = Y$
- *Symmetry:* $\Delta(X,Y) = \Delta(Y,X)$
- *Subadditivity (triangle inequality):* $\Delta(X,Z) \leq \Delta(X,Y) + \Delta(Y,Z)$

Question 1. Justify why it is the case that $\Delta(X,Y) = 0 \iff X = Y$.

Statistical distance lets us say when two (sequences of) distributions are "essentially the same," in an asymptotic sense.

Definition 1.4. Let $\mathcal{X} = \{X_n\}$ and $\mathcal{Y} = \{Y_n\}$ be sequences of probability distributions, called *ensembles*, indexed by security parameters $n \in \mathbb{N}$.[2] We say that $\mathcal{X}$ and $\mathcal{Y}$ are *statistically indistinguishable*, written $\mathcal{X} \overset{s}{\approx} \mathcal{Y}$, if

$$\Delta(X_n, Y_n) = \mathrm{negl}(n).$$

Example 1.5. Let X_n be the uniform distribution over $\{0,1\}^n$, and let Y_n be the uniform distribution over the nonzero strings $\{0,1\}^n \backslash \{0^n\}$. An optimal test A is the singleton set $A = \{0^n\}$, yielding $\Delta(X_n, Y_n) = 2^{-n} = \mathrm{negl}(n)$, so $\mathcal{X} \overset{s}{\approx} \mathcal{Y}$. (This can also be seen by calculating the summation in Lemma 1.1.) The analysis extends similarly to any Y_n that leaves out a $\mathrm{negl}(n)$ fraction of $\{0,1\}^n$. From this we can say that such ensembles $\mathcal{Y}$ are "essentially uniform," or *statistically pseudorandom*.

Can a statistically pseudorandom generator exist? This depends on the definition of "generator" (which we give in the next lecture), but for any meaningful definition of the term, it isn't possible! This can be shown by explicitly demonstrating an appropriate subset (or test) that distinguishes strings output by the generator from uniformly random ones; we will do this in the next lecture.

1.2 Computational Indistinguishability

We can define a natural analogue of statistical distance in the computational setting, where the "test" is implemented by an *efficient algorithm*. Namely, for distributions X and Y and an algorithm $\mathcal{A}$ (possibly randomized), define $\mathcal{A}$'s *distinguishing advantage* between X and Y as

$$\mathbf{Adv}_{X,Y}(\mathcal{A}) = |\Pr[\mathcal{A}(X) = 1] - \Pr[\mathcal{A}(Y) = 1]|.$$

[2] More generally, to accommodate restrictions on the allowed security parameters, both ensembles may be indexed by a common infinite subset of $\mathbb{N}$. In that case, all asymptotic statements refer to n tending to infinity within that subset.

(The output of $\mathcal{A}$ can be arbitrary, but we interpret 1 as a special output indicating that the test implemented by $\mathcal{A}$ is "satisfied," and any other output as "not satisfied.") For ensembles $\mathcal{X} = \{X_n\}$ and $\mathcal{Y} = \{Y_n\}$, we give the distinguisher the security parameter explicitly and define its advantage as the function

$$\mathbf{Adv}_{\mathcal{X},\mathcal{Y}}(\mathcal{A})(n) = |\Pr[\mathcal{A}(1^n, X_n) = 1] - \Pr[\mathcal{A}(1^n, Y_n) = 1]|.$$

We may suppress the input 1^n and the argument n when they are clear from context.

Definition 1.6. Let $\mathcal{X} = \{X_n\}$ and $\mathcal{Y} = \{Y_n\}$ be ensembles. We say that $\mathcal{X}$ and $\mathcal{Y}$ are *computationally indistinguishable*, written $\mathcal{X} \overset{c}{\approx} \mathcal{Y}$, if $\mathbf{Adv}_{\mathcal{X},\mathcal{Y}}(\mathcal{A}) = \mathrm{negl}(n)$ for every PPT algorithm $\mathcal{A}$.
If each X_n is a distribution over $\{0,1\}^{\ell(n)}$, we say that $\mathcal{X}$ is (computationally) *pseudorandom* if $\mathcal{X} \overset{c}{\approx} \{U_{\ell(n)}\}$, the ensemble of uniform distributions over $\{0,1\}^{\ell(n)}$.

The basic facts about statistical distance also carry over to computational indistinguishability, where we restrict all functions/tests to be *efficient*. We call the analogue of Lemma 1.2 the *composition lemma* (see Lemma 1.7), because it involves the composition of an arbitrary PPT procedure and distinguisher. We call the analogue of the triangle inequality the *hybrid lemma* (see Lemma 1.8), because we typically invoke it on a sequence of "hybrid" distributions that interpolate step-by-step between two distributions we ultimately care about.

1.2.1 Composition Lemma

Lemma 1.7 (Composition lemma). *Let $\mathcal{X} = \{X_n\}$ and $\mathcal{Y} = \{Y_n\}$ be ensembles, and let $\mathcal{S}$ be a PPT algorithm. Define $X'_n = \mathcal{S}(1^n, X_n)$ and $Y'_n = \mathcal{S}(1^n, Y_n)$, and let $\mathcal{X}' = \{X'_n\}$ and $\mathcal{Y}' = \{Y'_n\}$. For any distinguisher $\mathcal{D}'$ (regardless of efficiency), the composed distinguisher*

$$\mathcal{D}(1^n, \cdot) := \mathcal{D}'(1^n, \mathcal{S}(1^n, \cdot))$$

has advantage

$$\mathbf{Adv}_{\mathcal{X},\mathcal{Y}}(\mathcal{D})(n) = \mathbf{Adv}_{\mathcal{X}',\mathcal{Y}'}(\mathcal{D}')(n).$$

If $\mathcal{D}'$ is PPT, then so is $\mathcal{D}$. In particular, if $\mathcal{X} \overset{c}{\approx} \mathcal{Y}$, then $\mathcal{X}' \overset{c}{\approx} \mathcal{Y}'$.

Proof. The algorithm $\mathcal{D}$ runs $\mathcal{S}$ and then $\mathcal{D}'$, using independent randomness for the two runs, and outputs the answer of $\mathcal{D}'$. If $\mathcal{D}'$ is PPT, then so is $\mathcal{D}$, because $\mathcal{S}$ is PPT and its output length is bounded by its running time. By construction,

$$\begin{aligned}
\mathbf{Adv}_{\mathcal{X},\mathcal{Y}}(\mathcal{D})(n) &= |\Pr[\mathcal{D}(1^n, X_n) = 1] - \Pr[\mathcal{D}(1^n, Y_n) = 1]| \\
&= \left|\Pr[\mathcal{D}'(1^n, \mathcal{S}(1^n, X_n)) = 1] - \Pr[\mathcal{D}'(1^n, \mathcal{S}(1^n, Y_n)) = 1]\right| \\
&= \left|\Pr[\mathcal{D}'(1^n, X'_n) = 1] - \Pr[\mathcal{D}'(1^n, Y'_n) = 1]\right| \\
&= \mathbf{Adv}_{\mathcal{X}',\mathcal{Y}'}(\mathcal{D}')(n).
\end{aligned}$$

For the final claim, restrict to PPT distinguishers $\mathcal{D}'$. If $\mathcal{X} \overset{c}{\approx} \mathcal{Y}$, the left-hand side is negligible because $\mathcal{D}$ is PPT, so the right-hand side is negligible as well; since $\mathcal{D}'$ is an arbitrary PPT algorithm, $\mathcal{X}' \overset{c}{\approx} \mathcal{Y}'$. □

The algorithm $\mathcal{S}$ is often called a *simulator* because it "simulates" a sample from X'_n or Y'_n given a sample from X_n or Y_n, respectively. It is thus a *reduction* from distinguishing $\mathcal{X}$ and $\mathcal{Y}$ to distinguishing $\mathcal{X}'$ and $\mathcal{Y}'$.

Question 2. Let $\mathcal{X} = \{X_n\}$ be a pseudorandom ensemble. Is $\{\overline{X_n}\}$ necessarily pseudorandom as well, where $\bar{\cdot}$ denotes bitwise complement? *Hint:* use Lemma 1.7.

1.2.2 Hybrid Lemma

Lemma 1.8 (Hybrid lemma). *Let $X^0, \ldots, X^m$ be distributions for some positive integer m, and let $\mathcal{D}$ be any algorithm (regardless of efficiency). Writing $p_i = \Pr[\mathcal{D}(X^i) = 1]$, we have*

$$\mathbf{Adv}_{X^0, X^m}(\mathcal{D}) = \left| \sum_{i \in [m]} (p_{i-1} - p_i) \right| \leq \sum_{i \in [m]} \mathbf{Adv}_{X^{i-1}, X^i}(\mathcal{D}).$$

Proof. By telescoping and the triangle inequality,

$$\mathbf{Adv}_{X^0, X^m}(\mathcal{D}) = |p_0 - p_m| = \left| \sum_{i \in [m]} (p_{i-1} - p_i) \right| \leq \sum_{i \in [m]} |p_{i-1} - p_i| = \sum_{i \in [m]} \mathbf{Adv}_{X^{i-1}, X^i}(\mathcal{D}).$$

□

It is important to understand that even though the intended goal of $\mathcal{D}$ is to distinguish X^0 from X^m, the proof of Lemma 1.8 analyzes the same distinguisher on *every* hybrid distribution X^i. This is legitimate because $\mathcal{D}$ is just an algorithm, so we can analyze its behavior on any input distribution we like. If $\mathcal{D}$ has advantage ε in distinguishing between X^0 and X^m, the hybrid lemma guarantees that it has advantage at least ε/m in distinguishing between some adjacent pair X^{i-1} and X^i.

Corollary 1.9. *Let $\mathcal{X}^0, \mathcal{X}^1, \ldots, \mathcal{X}^m$ be ensembles, where m is a positive constant independent of n. If $\mathcal{X}^{i-1} \overset{c}{\approx} \mathcal{X}^i$ for every $i \in [m]$, then $\mathcal{X}^0 \overset{c}{\approx} \mathcal{X}^m$.*

Proof. Write $\mathcal{X}^i = \{X_n^i\}$ and fix any PPT distinguisher $\mathcal{D}$. For each security parameter n, apply Lemma 1.8 to $X_n^0, \ldots, X_n^m$ and the algorithm $\mathcal{D}(1^n, \cdot)$, obtaining

$$\mathbf{Adv}_{\mathcal{X}^0, \mathcal{X}^m}(\mathcal{D})(n) \leq \sum_{i \in [m]} \mathbf{Adv}_{\mathcal{X}^{i-1}, \mathcal{X}^i}(\mathcal{D})(n).$$

Each summand is a negligible function by hypothesis, and a sum of a constant number of negligible functions is itself negligible: for every constant $c > 0$, each summand is $O(n^{-c})$, so their sum is also $O(n^{-c})$. Since $\mathcal{D}$ was arbitrary, $\mathcal{X}^0 \overset{c}{\approx} \mathcal{X}^m$. □

Polynomially many hybrid steps. In many applications, the number of hybrid steps grows polynomially with the security parameter n. We cannot directly express this by replacing the constant m in Corollary 1.9 with a function $m(n)$, because n is in scope only within each ensemble. Instead, for each n, we define a sequence of distributions $H_n^0, \ldots, H_n^{m(n)}$; the initial and final distributions form the two ensembles $\mathcal{H}^0 := \{H_n^0\}$ and $\mathcal{H}^m := \{H_n^{m(n)}\}$ that we want to prove are indistinguishable. The hybrid bound (Lemma 1.8) applies to each such sequence, but bounding this sum by a negligible function requires additional care: the sum of polynomially many *different* negligible functions is not necessarily negligible.[3]

[3] For example, let $\nu_i(n) = 1$ when $n = i$ and $\nu_i(n) = 0$ otherwise. Each ν_i is negligible, but $\sum_{i=1}^{n} \nu_i(n) = 1$ for every n.

Corollary 1.10 (Polynomially many hybrid steps). *Let $m(n) \geq 1$ be integer-valued, with $1^{m(n)}$ polynomial-time computable from 1^n, so $m(n) = \mathrm{poly}(n)$. For each n, let $H_n^0, \ldots, H_n^{m(n)}$ be distributions, and define $\mathcal{H}^0 := \{H_n^0\}$ and $\mathcal{H}^m := \{H_n^{m(n)}\}$. Let $\mathcal{X} = \{X_n\}$ and $\mathcal{Y} = \{Y_n\}$ be computationally indistinguishable ensembles, and let $\mathcal{S}$ be a PPT algorithm. Writing $\mathcal{S}_i(1^n, z) := \mathcal{S}(1^n, i, z)$, suppose that for every n and every $i \in [m(n)]$,*

$$\mathcal{S}_i(1^n, X_n) \equiv H_n^{i-1} \qquad \text{and} \qquad \mathcal{S}_i(1^n, Y_n) \equiv H_n^i. \tag{1.1}$$

Then $\mathcal{H}^0 \overset{c}{\approx} \mathcal{H}^m$.

Proof. Define the *averaging simulator* $\bar{\mathcal{S}}$ that, on input $(1^n, z)$, chooses $i \leftarrow [m(n)]$ uniformly and runs $\mathcal{S}_i(1^n, z)$ with fresh randomness; clearly, this $\bar{\mathcal{S}}$ is PPT.

Fix any PPT algorithm $\mathcal{D}'$; we will show that it has negligible advantage in distinguishing $\mathcal{H}^0$ and $\mathcal{H}^m$. Let

$$p_i(n) = \Pr[\mathcal{D}'(1^n, H_n^i) = 1],$$

so the advantage is $|p_0(n) - p_{m(n)}(n)|$. To use the hypothesis $\mathcal{X} \overset{c}{\approx} \mathcal{Y}$, consider the composed PPT distinguisher $\mathcal{D}(1^n, \cdot) := \mathcal{D}'(1^n, \bar{\mathcal{S}}(1^n, \cdot))$ from Lemma 1.7. Unfolding the definition of $\mathcal{D}$, averaging over the choice of i using Equation (1.1), and telescoping the sum as in Lemma 1.8 gives

$$\begin{aligned}
\mathbf{Adv}_{\mathcal{X},\mathcal{Y}}(\mathcal{D})(n) &= \left|\Pr[\mathcal{D}'(1^n, \bar{\mathcal{S}}(1^n, X_n)) = 1] - \Pr[\mathcal{D}'(1^n, \bar{\mathcal{S}}(1^n, Y_n)) = 1]\right| \\
&= \frac{1}{m(n)} \left| \sum_{i=1}^{m(n)} \left(\Pr[\mathcal{D}'(1^n, H_n^{i-1}) = 1] - \Pr[\mathcal{D}'(1^n, H_n^i) = 1]\right) \right| \\
&= \frac{1}{m(n)} \left| \sum_{i=1}^{m(n)} \left(p_{i-1}(n) - p_i(n)\right) \right| \\
&= \frac{|p_0(n) - p_{m(n)}(n)|}{m(n)} = \frac{\mathbf{Adv}_{\mathcal{H}^0,\mathcal{H}^m}(\mathcal{D}')(n)}{m(n)}.
\end{aligned}$$

Since $\mathcal{X} \overset{c}{\approx} \mathcal{Y}$ and $\mathcal{D}$ is PPT, its advantage on the left-hand side is negligible. Multiplying by $m(n) = \mathrm{poly}(n)$ preserves negligibility, so the advantage of $\mathcal{D}'$ on the right-hand-side is negligible as well. Since $\mathcal{D}'$ was arbitrary, we finally conclude that $\mathcal{H}^0 \overset{c}{\approx} \mathcal{H}^m$, as claimed. □

What matters in Equation (1.1) is that the mapping of X_n, Y_n to the two adjacent hybrids has the same orientation for every i. That is, mapping X_n to H_n^i and Y_n to H_n^{i-1} for every i would work equally well: it simply negates the telescoping sum, leaving its absolute value unchanged. Either consistent orientation makes the acceptance-probability differences telescope *before* we take the absolute value. By contrast, averaging the absolute values of the differences $p_{i-1}(n) - p_i(n)$ would only *upper*-bound the advantage of $\mathcal{D}$ by the average advantage of $\mathcal{D}'$ between adjacent hybrids. Such a bound goes in the wrong direction: we need to upper-bound the advantage of $\mathcal{D}'$ between $\mathcal{H}^0$ and $\mathcal{H}^m$ in terms of the advantage of $\mathcal{D}$ between $\mathcal{X}$ and $\mathcal{Y}$, which we know is negligible.

1.2.3 Changing the Security Parameter

A cryptographic primitive evaluated at one security parameter may invoke another primitive at a different parameter. For example, in Lecture 3's hardness amplification, the direct product $f^{(m)}$ has input length $n' = n \cdot m(n)$, but evaluates f on blocks of length n. There we transferred a security guarantee expressed

in n to one expressed in n'. Here we ask the analogous question for computational indistinguishability: if two ensembles are indistinguishable at parameter n, do they remain so when indexed by n'?

Lemma 1.11 (Change of security parameter). *Let $I \subseteq \mathbb{N}$ be infinite, and let $\mathcal{X} = \{X_n\}_{n\in I}$ and $\mathcal{Y} = \{Y_n\}_{n\in I}$ be ensembles. Let $n' = n'(n)$ be an injective map from I to $\mathbb{N}$ such that $1^{n'}$ is computable in polynomial time from 1^n for $n \in I$, and let $I' = n'(I)$. Define $X'_{n'(n)} = X_n$ and $Y'_{n'(n)} = Y_n$, and let $\mathcal{X}' = \{X'_{n'}\}_{n'\in I'}$ and $\mathcal{Y}' = \{Y'_{n'}\}_{n'\in I'}$. If $\mathcal{X} \overset{c}{\approx} \mathcal{Y}$, then $\mathcal{X}' \overset{c}{\approx} \mathcal{Y}'$ with respect to the security parameter $n' \in I'$.*

Proof. Injectivity ensures that $\mathcal{X}', \mathcal{Y}'$ are well-defined. Fix any PPT distinguisher $\mathcal{D}'$ for the relabeled ensembles, and define

$$\mathcal{D}(1^n, z) := \mathcal{D}'(1^{n'(n)}, z).$$

Then $\mathcal{D}$ is PPT because $1^{n'(n)}$ is efficiently computable from 1^n. By construction, for every $n \in I$,

$$\mathbf{Adv}_{\mathcal{X},\mathcal{Y}}(\mathcal{D})(n) = \mathbf{Adv}_{\mathcal{X}',\mathcal{Y}'}(\mathcal{D}')(n'(n)).$$

Because $\mathcal{X} \overset{c}{\approx} \mathcal{Y}$, this common advantage is negligible in n for $n \in I$. Efficient computation of $1^{n'}$ from 1^n implies $n' = \mathrm{poly}(n)$, and injectivity implies $n'(n) \to \infty$ as $n \to \infty$ within I. The change-of-parameter property from Lecture 2 therefore makes the advantage negligible in n' for $n' \in I'$. Since $\mathcal{D}'$ was arbitrary, $\mathcal{X}' \overset{c}{\approx} \mathcal{Y}'$. □

Remark 1.12 (Non-uniform adversaries). The composition lemma (Lemma 1.7), hybrid lemma and its corollaries (Lemma 1.8 and Corollaries 1.9 and 1.10), and change-of-parameter lemma (Lemma 1.11) have non-uniform analogues. In each, the indistinguishability hypotheses and conclusions alike are against non-uniform PPT distinguishers, and the simulators may also be non-uniform. For composition, give the reduction both the simulator's and the distinguisher's advice; their combined length is still polynomial. In the hybrid arguments, use the same advice on every hybrid distribution, so the same probability calculations apply. For a change of parameter, the distinguisher's advice at n' has polynomial length in n because $n' = \mathrm{poly}(n)$.

Answers

Question 1. Justify why it is the case that $\Delta(X, Y) = 0 \iff X = Y$.

Answer. In the forward direction, because $\Delta(X, Y) = 0$ and all the terms in the sum from Lemma 1.1 are non-negative, we have that $|X(w) - Y(w)| = 0$ for all $w \in \Omega$, i.e., $X(w) = Y(w)$. Hence, $X = Y$. In the other direction, if $X = Y$, then by definition $X(A) - Y(A) = 0$ for all A, so $\Delta(X, Y) = \sup(\{0\}) = 0$.

Question 2. Let $\mathcal{X} = \{X_n\}$ be a pseudorandom ensemble. Is $\{\overline{X_n}\}$ necessarily pseudorandom as well, where $\bar{\cdot}$ denotes bitwise complement? *Hint:* use Lemma 1.7.

Answer. Yes: complementing a uniformly random string preserves its distribution. Let $\ell(n)$ denote the length of the strings distributed according to X_n. Define the efficient algorithm $\mathcal{S}(1^n, z) := \overline{z}$. Since $\mathcal{X} \overset{c}{\approx} \{U_{\ell(n)}\}$, Lemma 1.7 gives

$$\{\overline{X_n}\} = \{\mathcal{S}(1^n, X_n)\} \overset{c}{\approx} \{\mathcal{S}(1^n, U_{\ell(n)})\} = \{U_{\ell(n)}\},$$

where the last equality of distributions holds because bitwise complementation is a permutation of $\{0,1\}^{\ell(n)}$. Thus $\{\overline{X_n}\}$ is pseudorandom.